

# **MIRMIS/NIR–MIR Calibration pipeline description**

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Prepared by:

Valtteri Pitkänen

# Document Change Record

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|         |                     |                |                    |

## Applicable Documents

| AD    | Document Title                           |
|-------|------------------------------------------|
| AD-01 | MIRMIS Instrument Design Report          |
| AD-02 | Hera Image Format and Calibration Levels |
| AD-03 | ASP Calibration Report v0.81             |

## Reference Documents

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## Acronyms

|        |                                                          |
|--------|----------------------------------------------------------|
| ASPECT | Asteroid Spectral Imager                                 |
| AIS    | Asteroid Imaging Simulator                               |
| ESA    | European Space Agency                                    |
| FITS   | Flexible Image Transport System                          |
| FWHM   | Full Width Half Maximum                                  |
| HDU    | Header and Data Unit                                     |
| MIRMIS | Modular Infrared Molecules and Ices Sensor               |
| MIR    | Mid infrared single-point spectrometer channel on MIRMIS |
| NIR    | Near infrared hyperspectral imager channel on MIRMIS     |
| NASA   | National Aeronautics and Space Administration            |
| NOAA   | National Centers for Environmental Information           |
| PDS4   | Planetary Data System version 4                          |
| TIRI   | Thermal infrared imager channel on MIRMIS                |
| VTT    | VTT Finland                                              |

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# 1. Introduction

This document describes the preliminary design and implementation of the data processing pipeline for the MIRMIS instrument NIR and MIR channels. MIRMIS instrument package is being developed for the ESA Comet Interceptor mission. As part of the wider Comet Interceptor payload, it is developed to meet the mission's top-level science goal of understanding the diversity of comets by mapping the composition and temperature of the nucleus and coma [AD-01].

The MIRMIS instrument covers spectral range 0.9–25  $\mu\text{m}$  using three channels: NIR, hyperspectral imager (0.9–1.7  $\mu\text{m}$ ); MIR, single-point spectrometer (2.5–5.0  $\mu\text{m}$ ); and TIRI, multispectral thermal imager (6–25  $\mu\text{m}$ ) [RD4]. The MIR and NIR modules are based on tunable Fabry-Pérot interferometers which are used as an adjustable bandpass filter [RD10]. The NIR design is based on refractive optics with heritage from the ASPECT imager in the Milani CubeSat for ESA's Hera mission [RD5, RD6]. MIRMIS/NIR and MIRMIS/MIR are supplied by VTT Finland.

The MIRMIS/NIR–MIR data processing pipeline was developed to facilitate an accurate ground calibration of the acquired instrument data and generate ESA/NASA standard compliant data products (FITS/PDS4).

This document outlines the process and methodologies within the pipeline, which transforms raw instrument data into a single FITS file, containing a calibrated hyperspectral image of the target, complete with metadata about the acquisition. The result FITS file can be easily converted into a PDS4 product. Instructions to run the pipeline are found in the repository README.md file.

## 2. Overview

The pipeline is organized into distinct processing levels, each corresponding to a specific stage of the calibration or analysis. Each processing level consist of one or more steps. Beginning with raw data, each level progressively applies data calibration or processing procedures. Each processing level outputs its results in a FITS file. Level 0 creates a FITS file from the raw binary data, while level 1 focuses on calibrating and correcting the data. Both levels are applied individually for NIR and MIR channels.

The Figure 1 outlines processing levels applicable to each channel. Columns represents the individual imaging channels (NIR and MIR) while the rows correspond to a specific processing steps associated with a particular processing levels. A solid square within the diagram indicates that the corresponding step of the respective level is applicable to the particular channel. Each processing level builds upon the output of the preceding level. After each level (0A, 1A, 1B, and 1C) a new FITS file is created and saved.

### 2.1. Program and libraries

The data processing pipeline is developed with the Python programming language. Python is widely used in scientific computing and data processing due to its large ecosystem of libraries. Python provides libraries for data manipulation and image processing, which are essential for the tasks in this pipeline. The core libraries used in the pipeline are: Numpy (numerics), Astropy (FITS I/O and headers), SpicePy (SPICE data). For the full list of dependencies, see pyproject.toml in the repository.

### 2.2. Running the pipeline

The data from each acquisition is expected to arrive as binary files and JSON metafiles. Each exposure produces a separate binary file, while the associated metadata are provided in JSON format. The expected input directory structure is illustrated in Figure 2. The input directory is organized into two main subdirectories. The meta directory contains the metadata required to build the FITS headers and must include at least config.json and telemetry.json. The acquisition

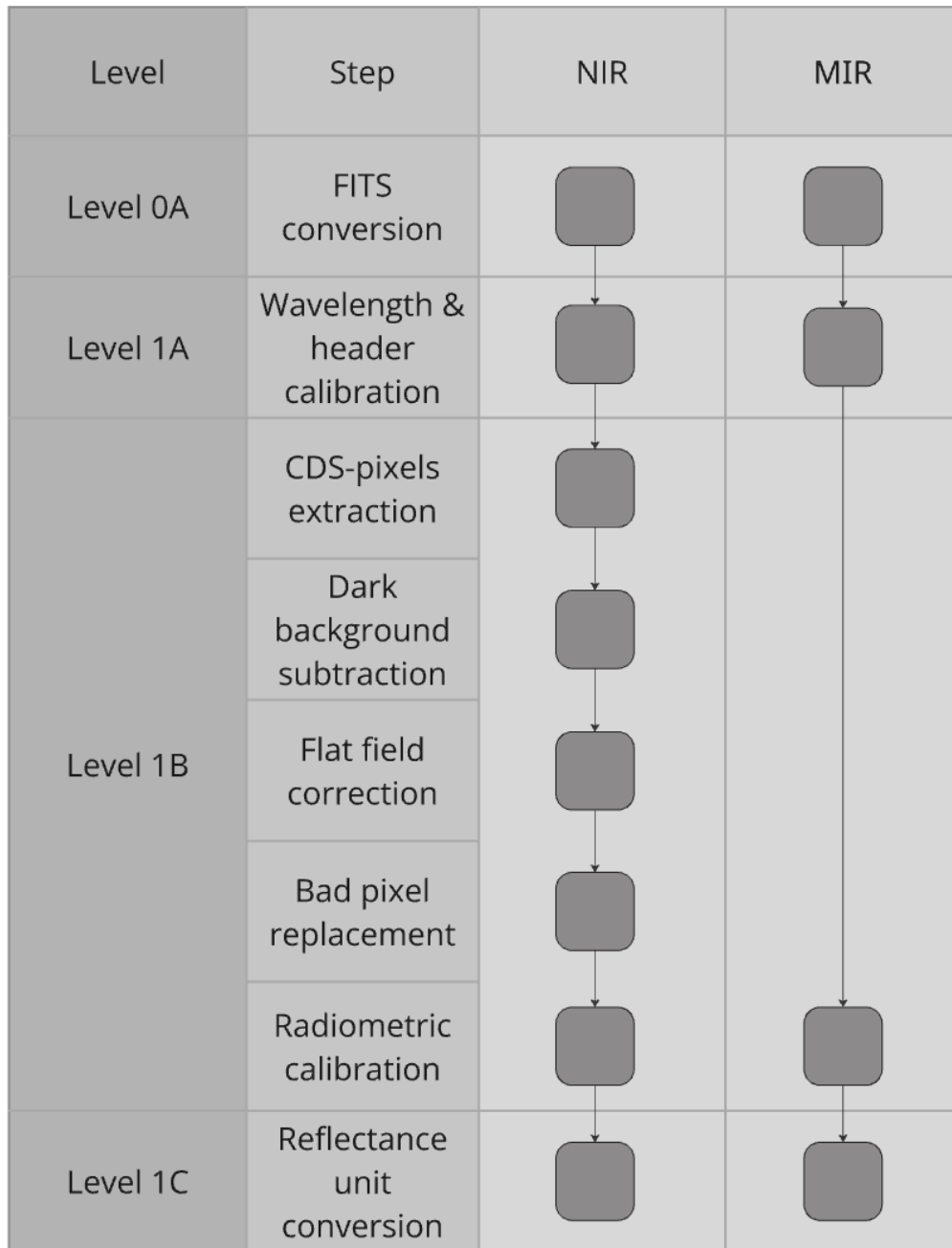


Figure 1: The pipeline with processing done in each data level. Columns represent the NIR and MIR channels and a square on particular data level indicates that corresponding procedure is applied to that channel.

data are stored in a directory whose name begins with 'acq\_';

Image files are stored in binary format and the expected naming convention is based on ASPECT data files naming defined in [AD-02]. Character 4 specifies the channel index—0 for NIR, and 1 for MIR. Characters 10-12 indicates the exposure number of that acquisition. For example, For



example, `dc_0_exp_000.bin` corresponds to the first frame of NIR channel. Files may appear either in raw binary format or as JPEG2000-decoded images with the `.jp2` suffix.

```

input_directory/
├── acq_000/
│   ├── dc_0_exp_000.bin.jp2
│   ├── dc_0_exp_001.bin.jp2
│   ├── dc_3_exp_000.bin
│   └── ...
├── meta/
│   ├── config.json
│   ├── telemetry.json
│   └── ...
└── meta_diff_encoding (optional)/
    └── diff_encodings.json

```

Figure 2: Input directory structure expected by the pipeline

The FITS file created at each processing level are stored into an output directory. The naming convention of the pipeline results and the structure of each processing level follow the guidelines defined in [AD-02] so the imaging channel and processing level can be easily identified from the FITS file name. Characters 1-4 specify the channel data included. Characters 5-10 represent an unique image number, increase by the acquisition time. It is generated as the base 32 coding of the image capture SC clock seconds. Characters 12-24 are the date and time of the exposure UTC: e.g. `230131T140112_` for UTC 2023-01-31T14:01:12. Characters 26-27 of the filename indicate the processing level. For example, `NIR_009FEQ_270323T060000_1A.fits` refers to a data product stored after completing the level 1A. Detailed instructions to run the pipeline are provided in the `README.md` file of the repository.

## 2.3. Data products

All FITS files created by the pipeline conform to the FITS standard version 4.0, as defined in [RD2]. FITS file structure consists of the mandatory primary header and data unit (HDU), followed by optional extensions or special records. In this pipeline, the main data are stored in the Primary HDU. This consisting of all relevant metadata, and a data unit containing either a three-dimensional hyperspectral data cube for NIR or an one-dimensional array for MIR derived from the corresponding raw acquisition binary files. For NIR, an additional Binary Table extension

is included from processing level 1B onward. This extension stores the CDS pixels extracted from the original near-infrared images [RD3].

### **2.3.1 Primary HDU**

The Primary HDU contains all metadata related to the image acquisition. This includes the information about the acquisition, relevant SPICE metadata, calibration-specific metadata, and the instrument channel specific metadata. In accordance with the FITS standard, the Primary header begins with a set of mandatory default entries, followed by the acquisition metadata entries.

### **2.3.2 Binary Table HDU**

Following the Level 1B processing the FITS files containing NIR data will include an additional Binary Table extension. This extension stores the CDS pixels that are extracted from each NIR frame.

## **3. Pipeline levels**

This section provides an overview of how each processing level transforms the NIR-MIR data. The format and data type of the data vary depending on the channel characteristics and the processing level applied as illustrated in the Figure 3. The amount of the image frames per instrument channel, denoted by  $n$  in the table, may also vary.

### **3.1. Level 0**

The level 0A converts raw instrument data into FITS format.

The level 0A combines the exposure frames of a single channel to a single data product stored in the Data Unit of the Primary HDU, with the metadata retrieved from the same input directory stored in the Primary Header.

| Processing level | NIR                              | MIR                   |
|------------------|----------------------------------|-----------------------|
| 0A               | (n x 518 x 648)<br>uint16        | (n,)<br>uint32        |
| 1A               | (n x 518 x 648)<br>uint16        | (n,)<br>uint32        |
| 1B               | (n x 512 x 640)<br>float32 (>f4) | (n,)<br>float32 (>f4) |
| 1C               | (n x 512 x 640)<br>float32 (>f4) | (n,)<br>float32 (>f4) |

Figure 3: The Data Unit shape and data type by processing level.

### 3.1.1 SPICE Metadata Retrieval

The SPICE Kernel Dataset contains ancillary information about various objects, spacecrafts, and instruments of the mission. Most notably it provides geometric and positional information of the target which can be queried based on e.g. time, frame, and observer [RD7]. The spice metadata is appended to the Primary header at the processing level 0A.

The metadata that is retrieved from SPICE includes spacecraft position, image target position and distance, sun position, solar distance and elongation, earth position and distance, spacecraft quaternions (angle), and SPICE version. The queries are implemented in Python with the usage of Spiceypy [RD1]. During the pipeline development of this pipeline, no meta kernel yet exist for Comet Interceptor mission, thus the part is inherited from the Hera mission ASPECT pipeline and likely some parts will need to be updated later.

### 3.1.2 Header

In accordance with the FITS standard, the Primary header begins with a set of mandatory default entries, followed by the acquisition metadata entries specified in the Table 1. Certain instrument metadata fields are specific to individual channels and exposures. These entries have `<channel>` and `<exp_id>` to indicate the corresponding channel and exposure number respectively.

The instrument-specific include the key temperature readings. The CCDTMP (detector temperature), FPI1 (spectral filter temperature 1), and FPI2 (spectral filter temperature 2) are provided for each channel, even if no other data from that channel is included in the data product. The channels present in the data product are listed as a comma-separated string in the 'CHANNELS' header entry.

Table 1: Primary header metadata entries.

| Key      | Description                                           | Type   |
|----------|-------------------------------------------------------|--------|
| INSTRUME | Camera ID, 'MIRMIS'                                   | String |
| ORIGIN   | 'ESA-COMET-INTERCEPTOR'                               | String |
| MISSPHAS | Mission Phase ID                                      | String |
| OSERV_ID | Observation ID                                        | String |
| FILENAME | Name of the FITS file                                 | String |
| ORIGFILE | Original filename, e.g. 'dc_X_exp_XXX.bin'            | String |
| SWCREATE | Software identification, e.g. 'NIR-MIRCAL v1.0'       | String |
| DATE     | UTC time of file creation, e.g. '2025-07-31T15:13:47' | String |
| PROCLEVL | Current calibration level                             | String |
| DATE-OB  | UTC time of observations                              | String |
| SC_CLK   | Spacecraft clock instrument format                    | String |
| OBJECT   | Observation object                                    | String |
| SPICE_MK | SPICE metakernel                                      | String |
| SPICEVER | SPICE dataset version                                 | String |

*Continued on next page*

Table 1: Primary header metadata entries.

| Key        | Description                                    | Type   |
|------------|------------------------------------------------|--------|
| SPICECLK   | SC clock SPICE format, e.g. '1/077436069.1856' | String |
| SUN_POSX   | Sun position vector X [km]                     | String |
| SUN_POSY   | Sun position vector Y [km]                     | String |
| SUN_POSZ   | Sun position vector Z [km]                     | String |
| SOLAR_D    | Solar distance [AU]                            | String |
| EARTPOSX   | Earth position vector X [km]                   | String |
| EARTPOSY   | Earth position vector Y [km]                   | String |
| EARTPOSZ   | Earth position vector Z [km]                   | String |
| EARTH_D    | Earth distance [AU]                            | String |
| TARGET     | Observation target (SPICE)                     | String |
| TRG_POSX   | Target position vector X [km]                  | String |
| TRG_POSY   | Target position vector Y [km]                  | String |
| TRG_POSZ   | Target position vector Z [km]                  | String |
| TRG_DIST   | Target distance [AU]                           | String |
| SC_QUATW   | Spacecraft quaternion (W)                      | String |
| SC_QUATX   | Spacecraft quaternion (X)                      | String |
| SC_QUATY   | Spacecraft quaternion (Y)                      | String |
| SC_QUATZ   | Spacecraft quaternion (Z)                      | String |
| CAM_RA     | Camera axis RA [deg]                           | String |
| CAM_DEC    | Camera axis DEC [deg]                          | String |
| CAM_NAZ    | Camera axis north azimuth [deg]                | String |
| SOL_ELNG   | Solar elongation                               | String |
| CHANNELS   | Channel data in this file, e.g. 'NIR'          | String |
| NIR_CCDTMP | NIR Detector temperature                       | String |
| NIR_FPI1   | NIR FPI 1 temperature                          | String |

*Continued on next page*

Table 1: Primary header metadata entries.

| Key                     | Description                                      | Type   |
|-------------------------|--------------------------------------------------|--------|
| NIR_FPI2                | NIR FPI 2 temperature                            | String |
| MIR_CCDTMP              | MIR Detector temperature                         | String |
| MIR_FPI1                | MIR FPI 1 temperature                            | String |
| MIR_FPI2                | MIR FPI 2 temperature                            | String |
| <channel>_FRAMES        | All frames included, comma-separated if multiple | String |
| <channel>_TASK_NUMBER   | Number of tasks                                  | String |
| <channel>_TASK_<exp_id> | 'sp1 sp2 sp3 exposure'                           | String |

## 3.2. Level 1

The purpose of level 1 is to calibrate and correct the data so that it accurately represents the information about the target in scientifically meaningful units. It ensures that the instrument's raw measurements are translated into values that can be reliably used in scientific analysis, reducing errors caused by instrument imperfections. The level is divided into 1A, 1B, and 1C. Level 1A converts metadata values like temperatures from DN values to kelvins and calculates the wavelengths from setpoint values. Level 1B applies multiple correction procedures to the data, namely, extracting diagnostic pixels, dark background subtraction, flat field correction, bad pixel replacement and radiometric calibration. Level 1C converts the radiance values into reflectance units.

### 3.2.1 Level 1A

The objective of header and wavelength calibration is to convert instrument temperature readings from raw DN values to calibrated temperatures in Kelvin, to convert the exposure DN values to seconds and to convert the piezo setpoint values into actual wavelength values (nm). Both channels utilizes a tunable Fabry-Pérot interferometer (FPI) developed at VTT Technical Research Center of Finland Ltd.. The FPI consist of two parallel reflective surfaces, which causes the light to reflect back and forth, creating a constructive and deconstructive interface. Essentially, the

FPI can be used as a bandpass filter, where a specific wavelength can be determined based on the tunable distance between the reflective surfaces. The distance is controlled in a closed loop, where the FPI control electronics measure the capacitance between three pairs of electrodes deposited on the mirror substrate [RD9]. The set point readings are stored into a task file from where the values can be converted into corresponding wavelengths by a mathematical formula 1, where  $sp$  is the set point value while  $A$  and  $B$  are constants determined during the calibration procedure. The value  $C$  is a correction value that can be modified based on wavelength calibration.

The converted wavelengths and exposures for each exposure are added to the header as individual entries. The naming convention for these header entries are described in Table 2. The wavelength for NIR exposure 000 would have the label 'NIR\_WL\_000'.

$$\lambda = A * (sp - C) - B \quad (1)$$

Table 2: Header entries added at level 1A

| Key                    | Description                             | Type   |
|------------------------|-----------------------------------------|--------|
| <channel>_WL_<exp_id>  | <channel> TASK <exp_id> wavelength [nm] | String |
| <channel>_EXP_<exp_id> | <channel> TASK <exp_id> exposure [s]    | String |

### 3.2.2 Level 1B

Level 1B processing modifies the data cube by calibrating the raw pixel values into radiance values. This level consist of several steps, each corresponding to a specific calibration procedure. All the calibrations are performed using double-precision values, and the final results are stored as 32-bit floating-point values to a FITS file. The applied calibration steps are described in Table 3.

Table 3: Calibration steps at Level 1B.

| Step                        | Description                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------|
| Float64 conversion          | The pixel values are converted to double precision for calculations                      |
| CDS-pixel extraction        | Extracts CDS pixels from NIR images are extracted and appended to Binary Table extension |
| Dark background subtraction | Remove dark current based on the master dark frame                                       |
| Flat field correction       | Flat field correction                                                                    |
| Bad pixels replacement      | Replace bad pixels by neighboring pixel average                                          |
| Radiometric calibration     | Scale the image pixels to radiance by the absolute calibration factor                    |
| Float32 conversion          | Convert pixel values to float32 for storing                                              |

### 3.2.3 CDS-pixel Extraction

In the original NIR images, the first six rows, the last row, and the four outermost columns on both sides consist of CDS pixels that do not form part of the actual image content [RD3]. These pixels are cropped and stored in a Binary Table extension. Each column in the table corresponds to a single frame and contains a one-dimensional array of 7,984 values, where the pixels are stored in row-major order, with pixels within each row listed from left to right. The column name encodes the channel index and frame number. For example, the column labeled NIR\_000 contains the CDS-pixels for frame 000 of the NIR channel. The cropping reduces the NIR image dimension from 648 x 518 pixels to 640 x 512 pixels.

### 3.2.4 Dark Background Subtraction

Dark background subtraction aims to remove unwanted background signals to enhance the accuracy of subsequent data interpretation. Ideally, an image acquired from a dark frame should consist solely of pixels with an intensity value of zero. However, in most sensors, dark frames include pixels with non-zero values due to dark current forming a background offset. This offset



adds to the light-generated signal during actual illuminated scene imaging. Background noise removal is achieved by subtracting a dark frame from the image. This process entails reducing each pixel value by the corresponding value in the dark frame. Mathematical representation of the dark background subtraction is given in equation 2, where  $B$  is the dark background mask.

$$I_{\text{corrected}}(x, y) = I_{\text{raw}}(x, y) - B(x, y) \quad (2)$$

### 3.2.5 Flat Field Calibration

Images generally exhibit a radial falloff of intensity from the center of the image. This effect is known as vignetting. The calibration process used to correct vignetting in images is called flat field correction. Flat field refers to a uniformly illuminated reference image, which records the sensor's response across all pixels. This reference image is used to adjust the intensity values by dividing each image with it. Mathematical representation of the flat field correction is given in equation 3, where  $F$  is the flat field mask.

$$I_{\text{corrected}}(x, y) = \frac{I_{\text{raw}}(x, y)}{F(x, y)} \quad (3)$$

### 3.2.6 Bad Pixel Removal

Bad pixel refers to a pixel that provides false information due to malfunctioning sensor. Identifying the bad pixels is part of the MIRMIS instrument calibration. The pipeline expects a bad pixel list as a .txt file. An example of the file structure is seen in table 4. The type column indicates the type of the problem: P for pixel, H for horizontal, V for vertical, and R for rectangular cluster. The second and third column indicates the location and the last two rows indicates the size of the affected area. Bad pixels are replaced by averaging the neighboring valid pixels in an iterative loop, such that all clusters are resolved.

Table 4: Bad pixel list example

| Type | Col | Row | SizeX | SizeY |
|------|-----|-----|-------|-------|
| P    | 300 | 200 | 1     | 1     |
| H    | -   | 0   | 640   | 1     |
| V    | 0   | -   | 1     | 640   |
| R    | 400 | 400 | 3     | 4     |

### 3.2.7 Radiometric Calibration

Radiometric calibration converts raw digital values into radiance units [ $\text{W m}^{-2} \text{sr}^{-1} \text{nm}^{-1}$ ] to ensure the accuracy and comparability of the data. To perform the conversion, conversion factors of type  $([\text{DN}/(\text{s} * \text{W m}^{-2} \text{sr}^{-1} \text{nm}^{-1})])$  are needed for each channel, (order) and wavelength. If an exact match for the wavelength is not found the pipeline estimates the conversion factor by linear interpolating values from the closest wavelengths on each side. The image DN values are then converted using the same formula 4, where  $I_{\text{exp}}$  is the exposure time in seconds and  $C$  is the conversion factor for the wavelength.

$$I_{\text{corrected}}(x, y) = \frac{I_{\text{raw}}(x, y)}{I_{\text{exp}} * C} \quad (4)$$

### 3.2.8 Level 1C

Level 1C converts the radiance values (result of 1B) to I/f reflectance units. The mathematical formula for this conversion is expressed in 5. The pipeline uses average solar spectral irradiance from year 2024, retrieved from NOAA archive [RD11]. The solar spectral irradiance is convolved with Gaussian spectral function ( $\text{FWHM} = 30 \text{ nm}$ ) and sampled at the instrument exposure wavelengths.

$$\frac{I}{F}(\lambda) = \frac{\pi L_{\lambda}(\lambda) r^2}{E_{\odot}(\lambda, 1 \text{ AU})} \quad (5)$$

In 5,  $\lambda$  denotes the wavelength,  $L_{\lambda}(\lambda)$  is the measured spectral radiance in units of  $\text{W m}^{-2} \text{nm}^{-1} \text{sr}^{-1}$ ,

and  $E_{\odot}(\lambda, 1 \text{ AU})$  is the solar spectral irradiance at 1 AU in units of  $\text{W m}^{-2} \text{ nm}^{-1}$ . The parameter  $r$  represents the heliocentric distance of the target expressed in astronomical units, with the factor  $r^2$  accounting for the inverse-square law governing solar illumination. The constant  $\pi$  arises from the definition of reflectance relative to a perfectly Lambertian surface. The resulting quantity  $I/F(\lambda)$  is the radiance factor, a dimensionless measure of reflectance that is normalized to the incident solar flux and independent of observation distance.

## 4. PDS4 Products

The Planetary Data System version 4 (PDS4) is the ESA/NASA standard for archiving and distributing planetary science data [RD8]. The pipeline supports generating PDS4-compliant data products from FITS files produced by the preceding processing levels, producing an XML label and a copy of the source FITS file for each data product.

The PDS4 labels are generated using a single shared Jinja2 [RD12] template that is populated with metadata extracted from the FITS file headers. The template is dynamically adapted to match the instrument channel (NIR or MIR) and the processing level of the data product.

The generated PDS4 labels are structured using the PDS4 common dictionary alongside four discipline dictionaries: Display, Imaging, Spectral, and Geometry. Once a mission-specific dictionary is defined for the Comet Interceptor mission, it should be incorporated into the labels. The label content and structure should also be coordinated with other instrument teams in the mission to ensure consistency across data products, and validated by the PDS node personnel once one is assigned to the mission.